SRILM Language Model Experiments

InClassActivity2 | Roll No. M25AI2151

Objective

The activity uses SRILM to count word sequences and build language models from the Brown corpus. Bigram and trigram models are evaluated on the test set, preprocessing variants are compared, and interpolation parameters are selected using the development set.

Method

The training text was prepared with sentence-boundary markers. A bigram count file and unsmoothed bigram model were created, followed by smoothed/Kneser–Ney models. Punctuation removal and punctuation-plus-stopword removal were evaluated separately. For interpolation, candidate weights and add-smoothing values were compared on the development set before evaluating a selected setting on test data.

Results

Model / experiment	Development-set selection	Test-set result
Unsmoothed bigram	Not tuned	18,639 OOVs; 93,612 zero-probability events; PPL 70.31; PPL1 96.46
Smoothed bigram (recorded run)	Not tuned	0 OOVs; 0 zero probabilities; PPL 642.33; PPL1 870.35
Kneser–Ney trigram	Not tuned	0 OOVs; 0 zero probabilities; PPL 340.18; PPL1 447.38
Unigram + bigram interpolation	Best tested $\lambda = 0.1$; development PPL1 = 427.84	0 OOVs; 0 zero probabilities; PPL 350.11; PPL1 461.05
Add-smoothing + interpolation grid	Best tested add-smoothing = 0.01, $\lambda = 0.3$; development PPL1 = 879.55	PPL 760.05; PPL1 1038.03

Preprocessing

Two preprocessing variants were run: (1) punctuation removed, and (2) punctuation and English stopwords removed. The resulting models and evaluation output were saved in *preprocessing-results.txt*. Exact comparative metrics are not reproduced here because they were not available in the notebook output inspected for this draft.

Brief observations

The unsmoothed bigram model assigned zero probability to many test events and left many words out of vocabulary. The recorded smoothed models avoided zero-probability events. The Kneser–Ney trigram run had lower test perplexity than the recorded smoothed bigram run. In interpolation, $\lambda = 0.1$ was best among the tested weights in the first run; the separate add-smoothing grid selected 0.01 and $\lambda = 0.3$. These two interpolation runs used different model settings, so their perplexities should not be treated as a direct controlled comparison.

Preliminary draft: verify the preprocessing log and confirm all required tasks against the activity sheet before finalizing.